DAVI KAZUHIRO NATUME

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OBJECTIVE

AI Engineering internship, building and integrating software solutions and machine learning models. Experience in fraud classification and detection, research in data stream mining, and development of systems with AI as a core engineering tool, from specification to documentation.

PROFESSIONAL EXPERIENCE

Software Development Intern July 2025 – Present
Bradesco Seguros *Curitiba, PR*

- I develop systems with AI as the main engineering tool, using Kiro to specify processes (BPMN, PDD and SDD), implement, and document solutions.
- Developed Java desktop applications to automate analysts' manual processes, reducing daily task execution time to less than 1 minute.
- Migrated VBA automations to Java, reducing the execution time of query routines by 80%.
- Developed a Java library based on the company's Design System to standardize the graphical interfaces of applications.
- Mentor new team members and analysts in programming logic and systems architecture.

SCIENTIFIC RESEARCH

Blockchain Fraud Detection with Data Stream Mining Algorithms July 2026 – August 2027
PIBIC PUCPR *Curitiba, PR*

- Hyperparameter tuning of machine learning classifiers for fraud detection, evaluating models by precision, recall, F1, and AUC.
- Study of benchmarks and ensemble techniques for data stream mining, with experiments in MOA (Massive Online Analysis).
- Development of the experimental protocol, including selection of blockchain fraud datasets and configuration of literature approaches.
- Comparison of data stream classification techniques in the context of blockchain fraud.

PROJECTS

Quarto Pixel: Interactive Pixel Art Portfolio October 2026 – Present
A game-style portfolio: an isometric pixel art room where each object opens a part of the professional journey. Code at github.com/davi-kazuhiro/portfolio.

- Development orchestrated with Claude Code: a main session splits each roadmap milestone into tasks and delegates them to five specialized agents (art, scene, interface, content, and quality), each restricted to its own area of the codebase.
- Use of MCPs for library documentation lookup (context7) and browser verification (Playwright).
- Architecture with typed communication between the game canvas and the interface through an EventBus, keeping layers decoupled. Decisions recorded in ADRs.
- Technologies: Phaser 4, React 19, TypeScript, Zustand, i18next (PT/EN), Vite, and Vitest.

Attendance System
Web application for class management and academic attendance tracking, specified and developed with Kiro, from the design system to the frontend and backend. MVC architecture with REST API, profile-based authentication (teacher and student), attendance recording with geolocation, and report generation. Technologies: React, JavaScript, Node.js, and MySQL.

EDUCATION

Bachelor's Degree in Computer Science Expected graduation: 2027
PUCPR – Curitiba, PR GPA (IRA): 9.4
Relevant coursework: Big Data, Statistics, Data Structures, Algorithm Complexity.

SKILLS

Machine Learning	Classification, ensembles, data stream mining, hyperparameter tuning, MOA
AI Engineering	Claude Code, Kiro, agents and agent orchestration, MCP, spec-driven development
Programming	Python, Java, Spring, TypeScript, JavaScript, React, Node.js, SQL, MySQL
Tools	Git, Linux, Vite, Vitest, Playwright, VSCode, Eclipse
Languages	Portuguese (native), English (intermediate-advanced)

EXTRACURRICULAR ACTIVITIES

- **SBC Programming Marathon Training:** preparation in competitive programming, focusing on algorithms, data structures, problem solving, and complexity analysis.